# WireClaim

THE COMPLETE VERSION — the two-page write-up is `writeup.pdf`

Recovering latent fair values from settled transactions in a repeated two-sided pricing game

Team **Bin busy** • QuantCo *Claim to Fame* (Track 1) • Munich Agentic Hackathon, 22–23 August 2026

**Abstract.** Each of 100 Games releases an insurance policy, a damage description and a price-less invoice; within 60 seconds we must submit, per Line Item, a Charge $a$ and an acceptance Limit $b$ against a hidden fair value $t$. We show (i) that the payoff table makes the Charge a *cliff* rather than a slope, so the optimal Charge sits strictly below the estimate; (ii) that $t$ is *exactly recoverable* after settlement from the published transaction log, which turns every design question into a measurement; and (iii) that under that instrument the whole remaining gap to optimal play is estimation error, not strategy. We report a climb from last place to 5th, the second-best per-Game rate in the field over the last twenty Games, and the second-best risk-adjusted return.

## 1 The game, and the row everybody skips

Per ordered pair (issuer $H$, reviewer $I$) and Line Item, with hidden fair value $t$ and hidden cap $c \geq \max(4t, 2000)$, the reviewer's payoff is

$$\pi_I = \begin{cases} -a & a \leq t,\ a \leq b \\ -\mathbf{1.5\,a} & a \leq t,\ a > b \\ -\min(a, c) & a > t,\ a \leq b \\ 0 & a > t,\ a > b \end{cases} \tag{1}$$

while the issuer receives $a$ whenever $a \leq t$ — *whatever the reviewer does* — and $\min(a, c)$ only if an Overcharge is accepted. So a fair Charge is owed by all $N = 16$ opponents; an Overcharge is paid by the few whose Limit happens to admit it. Measured on Game 62: **16 payers versus 3**. The Charge is a **cliff, not a slope**: one rival charged 22% less than us on a single Line Item, was paid by 16/16, and took 136,075 — 87% of their whole Game.

## 2 Both decisions fall out of (1)

**The Charge.** With $F$ the posterior CDF of $t$ and $q$ the fraction of opponents whose Limit admits $a$, income per opponent is $a[1 - F(a) + F(a)q(a)]$. Empirically $q$ collapses across $a = t$ (16/16 below, 17% at 1.0–1.3×, 7% above), and a *mis-measured* $q$ costs more than assuming $q = 0$, so we maximise the conservative objective. For $t \sim \mathrm{LogN}(\ln m, \sigma^2)$ and $a = k\,m$,

$$R(k) \ \propto\ k\,\Phi\left(-\tfrac{\ln k}{\sigma}\right), \qquad a = k(\sigma)\,m, \tag{2}$$

whose maximiser is *strictly below 1 at every precision* and falls as the estimate degrades (0.70 at $\sigma$=0.25, 0.60 at 0.45, 0.45 at 0.75). We ship $k(\sigma) = \mathrm{clamp}(0.85 - 0.45\sigma, 0.30, 0.80)$.

**The Limit.** By (1) accepting costs $a$; rejecting costs $1.5a$ *only if the claim was fair.* Accepting is therefore right exactly when $a < 1.5a\,P(t \geq a)$, i.e. $P(\text{fair}) > \frac{2}{3}$, so the Limit is the **one-third quantile of the posterior** — derived, not tuned. Coverage needs no branch: with $p = P(\text{covered})$ the posterior is a spike-and-slab,

$$P(t) = (1 - p)\,\delta_0 + p\,\mathrm{LogN}(\ln m, \sigma^2), \tag{3}$$

so when $p \leq \frac{2}{3}$ the bottom third *is* the atom at zero and $b = 0$ emerges on its own. 40% of settled Line Items have $t = 0$, and paying on one is a pure loss.

## 3 The model reads; the engine prices

No model output is ever a price. Models emit *structured evidence* — a coverage probability with the policy clause quoted verbatim, a price band, a quantity — and deterministic code turns evidence into the posterior and the posterior into $(a, b)$, combining channels by inverse-variance weighting in log space ($w_i = \sigma_i^{-2}$, $\sigma_{\text{memory}} = 0.43$ against $\sigma_{\text{model}} = 0.60$).

The reason is auditability under repetition: this ran unattended 100 times, once every 12.6 minutes, in a 60-second window. **A prompt that emits a number cannot be swept, folded or replayed; a prompt that emits evidence can.** It also degrades rather than fails — **Games 82–89, eight consecutive 3×-weighted Games, ran with the model endpoint returning 401 on every call** and still banked +254,092.

## 4 Why our counterfactuals are measurements

The published transaction log inverts. A *rejected* transaction carrying a non-zero amount was a wrongful rejection, which by row 2 of (1) occurs only when the Charge was fair; a rejection at zero occurs only when it was not. Hence

$$\boxed{t \in \Big[\max_{\text{rej, amt}>0} a_\alpha,\quad \min_{\text{rej, amt}=0} a_\alpha\Big)} \tag{4}$$

brackets $t$ for *every Line Item ever played.* Over 52,224 settled rows the reconstruction reproduces every published net **to the cent**, so we can replay any hypothetical submission against the real Field with all sixteen opponents held fixed. The replay's self-check asserts that our *actual* submission reproduces the authoritative net for every settled Game — a test, not a comment.

**The bar.** Positive on **all four folds** (odd, even, early, late), never merely in total, against a *measured* noise floor of 26,622 over 18 Games — ±6,275 for one Game, which is why no single Game ever justified a change. Of twenty numbered hypotheses, **twelve were measured and rejected**. One shipped and was **reverted twenty minutes later**: it passed a real calibration table and all four folds, but a mechanism check showed it moved **4 Line Items in 573**. A constant can be right about its data and wrong about its mechanism.

## 5 What we removed, and what we rejected

We also *removed* a clamp. Until Game 66 the code enforced $b \leq a$; since $b$ is a lower quantile of the same posterior $a$ is drawn from, this looks harmless and is not. Releasing it is worth +19,273 over 65 Games on all four folds and binds on 33% of items. Game 65 item 3 found it: our estimate was accurate to 1.4%, the clamp pinned $b = a = 199.01$, and thirteen *fair* Charges between 201 and 291 were all wrongfully rejected — −4,953 on one Line Item.

A representative selection of the twelve rejections, each with the script that killed it: a global Charge multiplier below 1.0 (monotone loss, −118,049 at ×0.9); the exact argmax of expected income (loses in **all 15 cells** of a two-parameter sweep); four separate attacks on the Limit (the derived $\frac{1}{3}$ is the argmax); reading the policy's exact EUR caps (the number is exact, the *row* is not identifiable, −5,905); more ensemble draws (draw noise is 1.1% of the error, a third draw buys 0.2%); and copying the best rival's $a/t$ and $b/t$ ratios, which applied to our own $\hat{t}$ is −280,000 to −380,000.

**Two traps, each of which cost a working session.** *Conditioning on the outcome:* bucketed by what items *turned out* to be worth, strictness pays 9.9:1 under 100 and loses above 500 — a compelling case for raising the Limit on expensive items. Bucketed by $\hat{t}$, the only split available at submission time, the gradient collapses: **of 23 items we believed were worth over 2,000, 14 were worth under half that**. *Censoring:* a Fair Value bracket with no upper bound is unbounded *because* nobody rightfully rejected, which correlates with the item being expensive — so every $\sigma$ we quote is optimistic, and we say so.

## 6 Results

| window | total | per Game | rate rank |
|---|---|---|---|
| G1–25 *(estimator not working)* | −322,595 | −12,904 | — |
| G26–94 *(since)* | +362,531 | +5,254 | **3 / 17** |
| G75–94 *(last twenty)* | +137,942 | +6,897 | **2 / 17** |
| season, weighted | +231,298 | — | **5 / 17** |

We were **17th of 17 — last — after nine Games**, and 5th from Game 76; the G1–25 deficit exceeds our entire season net, so the deficit *is* the learning curve. The result we would defend is the variance: **2 losing Games in the last twenty, worst** −3,941, and 4 in the last thirty, fewest of any team, for a mean-to-$\sigma$ of **0.72, second in the field**.

Season total asks how much money we ended up with, which for us is largely a question about Games 1–19: our cumulative net at Game 20 was −354,171, and the whole gap to the leaders is that hole rather than anything since. Rebasing every team to zero at Game 20, when our estimator went live, we are **2nd of 17** at +394,108 over the 74 Games since. At the conservative Game 26 anchor we are 3rd, with the top three inside 9,702 of each other — and our per-Game rate is marginally *higher* from 26 than from 20, so the earlier anchor is not the flattering one it looks like.

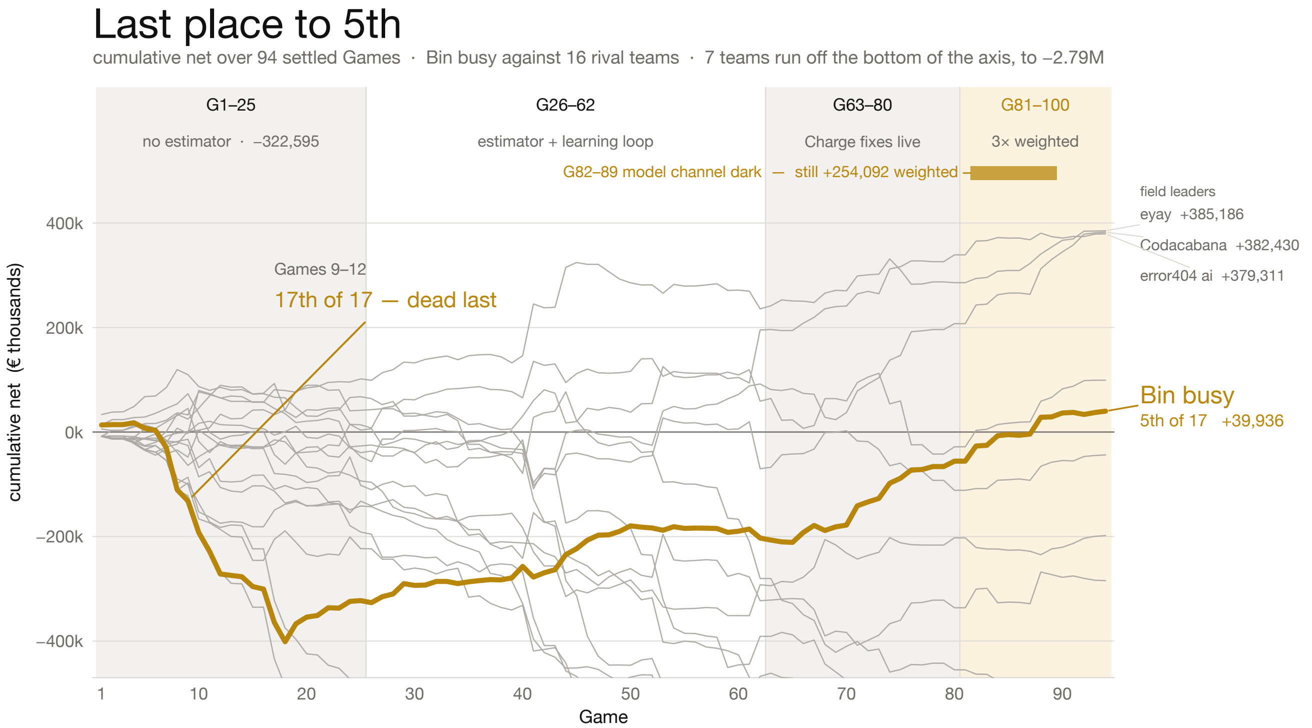


**Figure 1:** Cumulative net over 94 settled Games, all 17 teams; we are the heavy line. The deficit is Games 1–25, before the estimator was working; the slope after it is the method. Games 82–89 ran with the model channel returning 401 on every call.

## Why we think we succeeded — and where we did not

### Succeeded

- **We built the measuring instrument before the strategy.** Equation (4) recovers $t$ exactly, so every later decision was settled by replaying it against the real Field rather than argued. Three claims we had written down as fact were falsified within a day.
- **Last place to 5th.** 17th of 17 after nine Games; 5th from Game 76. Second in the field by rate over the last twenty Games; third of seventeen over the 69 Games since the estimator was working, within 4.3% of the leader.
- **The distribution is tight, and that is the result we would defend.** Two losing Games in the last twenty, the worst costing 3,941; four in the last thirty, fewest of any team. Mean-to-$\sigma$ of 0.72, second best in the field, while seven teams carry single Games worse than −80,000. In an insurance book the narrow distribution is the number that matters.
- **It degraded instead of failing.** Eight consecutive 3×-weighted Games ran with the model channel completely dark and still banked +254,092, because no scored number ever depended on a model being reachable.
- **We priced uptime correctly.** Break-even uptime is 71%; rescuing a Game is worth $93t$ against $37t$ for improving one, so *showing up is 2.5× being right.* A blind floor posts at T+0 before the Case is decrypted, submissions merge per Line Item, and a supervisor restarts on any crash. Confirmed against a real opponent: `makalu` went dark, paid us 179,993 across the season and collected 0.00 from anyone.

### Did not succeed

- **We spent the first quarter of the tournament building the instrument instead of scoring.** Games 1–25 cost −322,595 — more than our entire current season net. The rate says the decision was right; the total says we paid full price for it.
- **The estimate, not the strategy, is the whole remaining gap — and we learned that late.** $a = b = t$ reaches 100.3% of optimal play; $a = b = \hat{t}$ scores −50,140. The 2,498,118 between them is pure estimation error. We tuned decision rules to their measured argmax and it barely mattered; arguably we worked one layer too low.
- **We diagnosed our largest open lever and deliberately did not ship it.** Policy clause 7.1.5 has two halves and the model reads only the first. Game 74 returned coverage 0.01–0.05 on 20 of 31 Line Items, for 87% of that Game's penalties. Perfect coverage is worth +1,173/Game. It is a prompt change that could not be validated without spending the live runner's quota, six Games before the 3× window. We think the call was right and the outcome is still a loss.
- **Eight triple-weighted Games ran with the model dark and we did not notice.** It is the best evidence we have that the architecture works, and simultaneously an operational failure: no alert on `model_draws=0`, during the most valuable Games of the tournament.
- **We were beaten on the thing we understood best.** We had written down that the Charge is a cliff and still sat above $t$ too often; a rival took 136,075 off a single Line Item by charging 22% lower than us. Knowing a mechanism and being calibrated to it are different achievements, and we only closed that gap at Game 63.

**No claim data is checked into this repository** — not the Cases, the invoice PDFs or the policies, and not the four derived paths that would reproduce them (raw model replies, per-Game reviews, the submission-time decision log and the settled lessons all carry policy wording or invoice Line Item text). All four regenerate locally and are git-ignored. Single-clause quotations appear where the argument needs them, as citation rather than as an archive. Every figure in this document is reproducible from a script in the repository.